

Andrei Kozyrev

Curriculum Vitae

+49 176 84270634
✉ kozyrev.andrei.a@gmail.com
🌐 andrei-kozyrev.net

Education

- 2020–2022 **The Faculty of Mathematics and Computer Science, Modern Software Engineering,** *Saint-Petersburg State University*, Saint-Petersburg, Russia
- 2022–2023 **Bachelor of Science, Computer Science,** *Constructor University Bremen*, Germany, Bachelor of Computer Science. GPA: **B+**
- 2023–2025 **Master's of Science, Advanced Software Technology,** *Constructor University Bremen*, Germany

Publications

- arXiv'26 **A. Kozyrev, N. Khramov, D. Lochmelis, V. Morelli, G. Solovev, A. Podkopaev,** *RocqSmith: Can Automatic Optimization Forge Better Proof Agents?*
- AAMAS'26 **A. Kozyrev, N. Khramov, G. Solovev, A. Podkopaev,** *RocqStar: Leveraging Similarity-driven Retrieval and Agentic Systems for Rocq generation*
- ASE'24 **A. Kozyrev, G. Solovev, N. Khramov, A. Podkopaev,** *CoqPilot: LLM-based Proof Generation in Coq*

Professional Experience

- 2023–present **Research Scientist, JetBrains Research,** *Bremen*, Germany
Proof automatization using Language Models. AI Agents and their optimization.
- 2022–2023 **Intern at JetBrains Programming Languages Lab,** *Bremen*, Germany
Proof automatization for Coq proof assistant using OCaml & Rust.
- 2022 **Swift Developer,** *Saint-Petersburg*, Russia
IOS Software Developer in an outsourcing company, Spring 2022.

Selected Projects

A note, *names are referal to github projects*

- 2025 **Rocq RAG,** *A novel approach to Retrieval Augmented generation in Rocq language. Developed during work in JetBrains-Research, Rocq (Coq), PyTorch, TypeScript*
- 2024 **CoqPilot,** *A Visual Studio Code extension that is designed to help automate writing of Coq proofs. Developed while working in JetBrains-Research, started as a one-person project, and has grown to a three-people team. Repository on GitHub has more than 100 stars, paper is published on ASE'24, TypeScript, Coq*
- 2023 **Proofs of relational expressions via EGG,** *Equality saturation for solving equalities of relational expressions, bachelor thesis in Constructor University Bremen, Rust, OCaml, Coq*
- 2022 **High-Performance Network Programming in Swift,** *Research on Highly-efficient network programming using Swift and any competitive ability of Swift in this domain, Swift, Java, Python*
- 2022 **Progressive GAN Pytorch,** *Progressive generative adversarial network implementation from "Progressive growing of GANs for improved Quality, Stability, and Variation" paper, Python*

Contests

- Place 16 globally in **Reply Code Challenge** 2021.
- Place 407 globally in **Google Hash Code** 2021.